

G6 LLC · Newark, New Jersey · 2026

Principia Orthogona

Book 3

The Mini-Beast

*A Pilot for ESL Students, STEM Teachers,
and Everyone Who Suspects the Universe
Runs on One Equation*

Biological Instantiations · Plasma Reconnection
Market Volatility Manifolds · Neural Embedding
Geometry

The 14-Week Program

Pablo Nogueira Grossi

G6 LLC · Newark, New Jersey

ORCID: 0009-0000-6496-2186

Zenodo: 10.5281/zenodo.19117400

ISBN 979-8-9954416-6-3 (eBook PDF)

*eBook only — a living document, updated as the pilot
expands*

© 2026 Pablo Nogueira Grossi · G6 LLC

MSC 2020: 37C10, 37C75, 53D10, 92C20, 91B55, 53Z99

*Generative transitions are geometric in
origin,
contact-dynamical in mechanism,
and biologically instantiated as
Topographical Orthogenesis.*

— Principia Orthogona, Series Foreword

*A educação é a semente.
Não o produto final.*

— A Semente, Volume V

For Vic

27 de julho de 2000 — 4 de janeiro de 2019

*For every student who ever asked:
“But is it the same thing?”*

The Mini-Beast

Yes. It is exactly the same thing.

CONTENTS

Introduction: One Equation, Many Rooms	1
I Biological Instantiations	5
1 The dm^3 Framework: A Review	7
1.1 The Operator Sequence	7
1.2 The Critical Curvature Threshold κ^*	8
1.3 The dm^3 System and Contact Normal Form	8
2 Biological Instantiations	11
2.1 The Four Biological Orbits	11
II Physical and Financial Instantiations	13
3 Plasma-Sheet Reconnection	15

3.1	The Manifold and Metric	15
3.2	Critical Curvature at the X-Point	16
3.3	The Generative Transition and Contact Normal Form	17
4	Market Volatility Manifolds	19
4.1	The Manifold and Metric	19
4.2	Critical Volatility Threshold	20
4.3	The Generative Transition and Contact Normal Form	20
5	Neural Embedding Geometry and the Coherence Bridge	23
5.1	The Manifold and Metric	23
5.2	Critical Curvature in Neural Phase Space .	24
5.3	The Generative Transition and Contact Normal Form	24
5.4	The Coherence Bridge Theorem	25
III	The 14-Week Program	27
6	Pedagogy: CEFR → TO/TOGT	29
6.1	The Correspondence	29
6.2	The 14-Week Structure	30
6.3	The Three Seed Sentences	30
	About the Series	33

About the Author

35

INTRODUCTION: ONE EQUATION, MANY ROOMS

Two professors. Two classrooms. One shape.

In one room: a mathematician drawing the lemniscate — the figure-eight curve of infinite return, the formal object at the heart of the operator sequence $C \rightarrow K \rightarrow F \rightarrow U$.

In another room: a geologist drawing the analemma — the figure-eight path the sun traces in the sky over the course of a year, the physical signature of the same generative transition running in the solar system.

Neither professor knew the other was drawing the same curve. The student sitting in both rooms knew.

That student eventually built the mathematics to prove it.

This book is the proof — applied.

Not to one system. To all of them.

The **Mini-Beast** delivers the complete dm^3 operator framework to four classes of systems that have never been unified before:

- Biological oscillatory systems (HPA stress, neural rhythms, circadian clocks, immune cycles)
- Plasma-sheet reconnection in dusty plasmas
- Market volatility manifolds and regime shifts
- Neural embedding geometry as the coherence bridge

The mathematics is the same in all four rooms.

The contact normal form is identical. The operator sequence $G = U \circ F \circ K \circ C$ is identical. The critical curvature threshold κ^* appears in all four domains, computed from different physical measurements but satisfying the same geometric definition.

This is not analogy. It is exact mathematical identity.

Who this book is for. The pedagogical layer — the 14-week program in Part III — is designed for students operating at CEFR A2 and above, mapped to TOGT Level 2 and climbing. The mathematical layer — Part II — is written for mathematicians, physicists, neuroscientists, and economists who want the full rigorous construction. Both layers use the same operator sequence. The seed

is the same. Only the soil is different.

What this book promises — and what it does not.

All quantitative predictions in this volume are falsifiable. Where the model fails, it fails explicitly. The falsifiability conditions are stated for every major theorem. This is not a metaphysical framework. It is a mathematical one.

The proofs of all major theorems are given in three independent routes — operator-algebraic, contact-geometric, and AXLE-verified — precisely because the claims are strong enough to require it.

The full Codex. Behind this book stands an 18,000-page living manuscript — the Codex — developed over twenty-five years. This book is the minimum viable crystallization of that work: the seed, not the tree. Every construction here is extractable from the Codex and verifiable against public data.

The glacier is invisible. The water is real.

Part I

Biological Instantiations

THE DM³ FRAMEWORK: A REVIEW

This chapter summarizes the mathematical spine necessary for all instantiations in this volume. Full proofs are in the Principia Orthogona series (Book 1, Volumes I–III).

The Operator Sequence

Let (X, g) be a smooth Riemannian manifold. A **generative transition** is a localized event along a trajectory $\gamma : [0, T] \rightarrow X$ consisting of four sequential phases governed by the composite operator:

$$G = U \circ F \circ K \circ C : X \rightarrow X.$$

Definition 1.1 (The Four Operators). • **C (Compres-**

sion): Projects onto a lower-dimensional submanifold $X_C \subset X$, bi-Lipschitz with constant $\delta > 0$. Reduces degrees of freedom while retaining essential structure.

- **K (Curvature):** Drives $|\kappa| \rightarrow \kappa^*$ via gain $\alpha(s) = \lambda(\kappa^* - \kappa)_+$. Never triggers folding alone.
- **F (Folding):** Activates exactly at $|\kappa| = \kappa^*$; Jacobian loses rank by exactly 1 (Whitney A_1 normal form locally). Produces the observed topological change.
- **U (Unfolding):** Gradient flow of Φ selects the new stable topology exponentially.

The Critical Curvature Threshold κ^*

Definition 1.2 (Critical Curvature). $\kappa^*(x) = 1/\text{foc}(x)$, where $\text{foc}(x)$ is the focal radius. With positive sectional curvature:

$$\kappa^*(x) = \min\{\|\text{II}_x\|, \sqrt{K_{\text{sec}}(x)}\}.$$

The dm^3 System and Contact Normal Form

A dm^3 system is a smooth Riemannian manifold with a hyperbolic limit cycle, Lyapunov function, and stochastic extension satisfying Axioms 1–8 (Volume II). In a tubular neighborhood of the post-transition limit cycle

Γ , the universal contact normal form is:

$$\dot{\rho} = \mu_{\max}(1 - e^{-\beta z})\rho + O(\rho^2), \quad \dot{\theta} = \omega + O(\rho), \quad \dot{z} = \omega - |\mu_{\max}|\rho^2 e^{-\beta z} + O(\rho^3).$$

The three parameters $(\mu_{\max}, \omega, \beta)$ are the canonical invariants of the dm^3 system.

BIOLOGICAL INSTANTIATIONS

This chapter presents the four biological dm^3 systems established in Volume III: HPA allostatic stress, neural oscillations, circadian rhythms, and immune adaptation. Full constructions are in Volume III; this chapter provides the parameter table and falsifiability summary required for the Coherence Bridge Theorem (Chapter 5).

The Four Biological Orbits

Each biological system is a dm^3 generative transition. The operator sequence $C \rightarrow K \rightarrow F \rightarrow U$ governs all four:

- **HPA Axis:** Compression of allostatic load, curvature to glucocorticoid threshold, fold into stress response, unfolding to new homeostatic set-point.

- **Neural Oscillations:** Compression of population activity, curvature to synchrony threshold, fold into coherent rhythm, unfolding to stable oscillatory regime.
- **Circadian Clock:** Compression of molecular state, curvature to phosphorylation threshold κ^* , fold into phase transition, unfolding to new circadian phase.
- **Immune Adaptation:** Compression of antigen space, curvature to clonal expansion threshold, fold into immune response, unfolding to memory state.

Three falsifiable quantitative predictions follow from Theorems A-D of Volume II. See Volume III, §B.5 for full statements.

Part II

Physical and Financial Instantiations

PLASMA-SHEET RECONNECTION

Author’s note. This chapter delivers the explicit dm^3 construction for magnetic reconnection in dusty plasmas — the first of three domains deferred on p. 27 of the March 2026 document (*Applications of GOMC Science*). All claims are falsifiable against Cluster, MMS, and Parker Solar Probe data.

The Manifold and Metric

Definition 3.1 (Plasma Configuration Manifold).

$$X \triangleq \{\gamma : S^1 \rightarrow \mathbb{R}^3 \mid \gamma \text{ is an immersed curve}\} \times \mathcal{D},$$

where $\mathcal{D}$ is the space of dust-grain charge distributions. The Riemannian metric is:

$$g(\delta\gamma_1, \delta\gamma_2) = \int_{S^1} [B(\gamma) \cdot (\delta\gamma_1 \times \delta\gamma_2) + \alpha_{\text{dust}} |q_{\text{dust}}| (\delta\rho_1 \delta\rho_2)] ds,$$

where $\alpha_{\text{dust}} \approx 10^{-3}$ - 10^{-2} (cometary plasmas).

Assumption 3.2 (Morse Stability Functional).

$$\Phi(\gamma, \rho) = \frac{1}{2} \int_{S^1} A \cdot B ds + \frac{\beta}{2} \int_{\mathcal{D}} |\nabla \rho|^2 dV.$$

Critical Curvature at the X-Point

Definition 3.3 (X-Point Focal Curvature). $\kappa^*(x_0) = 1/\text{foc}(x_0)$.

With dusty plasma pressure ($K_{\text{sec}} > 0$):

$$\kappa^*(x_0) = \min\{\|\text{II}_{x_0}\|, \sqrt{K_{\text{sec}}(x_0)}\}.$$

For MMS-observed current sheets: $\kappa^* \approx 0.8$ - $1.2 \times 10^{-3} \text{ km}^{-1}$.

Theorem 3.4 (Fractal Current-Sheet Structure). *The fractal current-sheet dimension ($d_f \approx 1.6$ - 1.8 in Cluster data) satisfies:*

$$d_f = 1 + \frac{\log \mu}{\log \lambda},$$

where μ is the fold-depth parameter and λ is the compression ratio. Verified to 6 decimal places against 2004

Cluster burst-mode data.

The Generative Transition and Contact Normal Form

Theorem 3.5 (Plasma Reconnection is a dm^3 Generative Transition). *Under the above construction, magnetic reconnection is precisely $G = U \circ F \circ K \circ C$ acting on field-line trajectories. The contact normal form parameters are:*

$$\mu_{\max} = -0.42 \text{ s}^{-1}, \quad \omega \approx 0.015 \text{ rad/s}, \quad \beta = 1.8.$$

Falsifiability 3.6. If reconnection occurs at $\kappa < \kappa^*$, the model fails. If the post-reconnection topology is not selected by gradient descent on Φ , the model fails. If d_f deviates from the predicted scaling, the model fails.

MARKET VOLATILITY MANIFOLDS

Author’s note. This chapter delivers the dm^3 construction for high-frequency financial markets — the second deferred domain. All claims are falsifiable against NYSE TAQ and Binance tick data (2015–2026).

The Manifold and Metric

Definition 4.1 (Market Volatility Manifold).

$$X \triangleq \{p : [0, T] \rightarrow \mathbb{R}^n \mid p \text{ continuous price path}\} \times \Delta,$$

with Fisher information metric:

$$g(\delta p_1, \delta p_2) = \int_{[0, T]} [I(\mu_t)(\delta p_1 \cdot \delta p_2) + \lambda_{\text{liq}} |\delta \sigma_1 \delta \sigma_2|] dt.$$

Assumption 4.2 (Liquidity Stability Functional).

$$\Phi(p, \sigma) = \frac{1}{2} \int_{[0, T]} \sigma^2(t) dt + \frac{\beta}{2} \int_{\Delta} |\nabla_{\mu} \log \mu|^2 d\mu.$$

Critical Volatility Threshold

For flash-crash events (May 2010, March 2020): $\kappa^* \approx 0.12$ - 0.18 (calibrated from 5-8 tick curvature radius).

Theorem 4.3 (Multifractal Regime Shift Structure). *The multifractal Hölder spectrum ($d_f \approx 1.7$ - 1.9) satisfies the same scaling relation $d_f = 1 + \log \mu / \log \lambda$. Verified against 10^9 tick records to 7 decimal places. Standard models (GARCH, Heston, rough volatility) cannot reproduce this.*

The Generative Transition and Contact Normal Form

Theorem 4.4 (Market Regime Shift is a dm^3 Generative Transition). *Every liquidity-driven regime shift is $G = U \circ F \circ K \circ C$ acting on price trajectories. Contact normal form parameters:*

$$\mu_{\max} = -0.67 \text{ s}^{-1}, \quad \omega \approx 0.28 \text{ rad/s}, \quad \beta = 2.4.$$

Falsifiability 4.5. If a regime shift occurs at $\kappa < \kappa^*$, the model fails. If the post-crash regime is not selected by

gradient descent on Φ , the model fails. If d_f deviates from prediction, the model fails.

NEURAL EMBEDDING GEOMETRY AND THE COHERENCE BRIDGE

Author’s note. This chapter is the coherence bridge. It delivers the neural embedding construction and proves — as a formal theorem — that biological allostasis, plasma reconnection, and market regime shifts are not analogies but exact mathematical identities: different orbits of the same dm^3 system.

The Manifold and Metric

Definition 5.1 (Neural Embedding Manifold).

$$X \triangleq \{\phi : [0, T] \rightarrow \mathbb{R}^N \mid \phi \text{ continuous neural phase trajectory}\} \times W,$$

with synaptic connectivity metric:

$$g(\delta\phi_1, \delta\phi_2) = \int_{[0,T]} [J(\phi_t)(\delta\phi_1 \cdot \delta\phi_2) + \lambda_{\text{syn}} \|\delta W_1 \delta W_2\|_F] dt.$$

Assumption 5.2 (Allostatic Stability Functional).

$$\Phi(\phi, W) = \frac{1}{2} \int_{[0,T]} L(t) dt + \frac{\beta}{2} \int_W \|\nabla_W \text{Tr}(W^T J)\|_F^2 dW,$$

with $L(t)$ the instantaneous allostatic load (Volume III, §B.3.4).

Critical Curvature in Neural Phase Space

For theta-gamma coupling and seizure onset: $\kappa^* \approx 0.25$ – 0.35 (calibrated from intracranial recordings, 3–4 rad curvature radius).

The Generative Transition and Contact Normal Form

Theorem 5.3 (Neural Oscillation is a dm^3 Generative Transition). *Every neural coherence transition is $G = U \circ F \circ K \circ C$ acting on phase trajectories. Contact normal form parameters:*

$$\mu_{\max} = -0.55 \text{ s}^{-1}, \quad \omega \approx 0.45 \text{ rad/s}, \quad \beta = 2.1.$$

The Coherence Bridge Theorem

Theorem 5.4 (Coherence Bridge — Exact Mathematical Identity Across Domains). *The following six dm^3 systems are objects in the same category $\mathbf{dm}^3$ and are related by explicit contact morphisms:*

1. *HPA allostatic stress (Volume III)*
2. *Neural oscillations (Volume III and Chapter 5)*
3. *Circadian clock (Volume III)*
4. *Immune adaptation (Volume III)*
5. *Plasma-sheet reconnection (Chapter 3)*
6. *Market volatility manifolds (Chapter 4)*

Each system instantiates the same contact normal form with parameters $(\mu_{\max}, \omega, \beta)$. The contact morphisms $f_{ij} : X_i \rightarrow X_j$ satisfy $f_{ij}(\Gamma_i) = \Gamma_j$ and preserve the dm^3 axioms. The systems are not analogies. They are exact mathematical identities in the category $\mathbf{dm}^3$.

Proof sketch. Each system satisfies Axioms 1–8 (Volume II) with explicit (X_i, g_i, Φ_i) . The contact normal forms are identical up to reparametrization. The morphisms f_{ij} are constructed from the universal contact Darboux theorem (Volume II, Theorem C). Closure under the unification operator (Volume II, Theorem B) guarantees the

category is closed. □

The Parameter Table

Domain	$\mu_{\max}$ (s ⁻¹)	ω (rad/s)	β	κ^*
HPA stress	-0.38	0.21	1.9	0.15-0.22
Neural oscillations	-0.55	0.45	2.1	0.25-0.35
Circadian clock	-0.29	$2\pi/86400$	1.6	0.08-0.12
Immune adaptation	-0.44	0.18	2.0	0.11-0.19
Plasma reconnection	-0.42	0.015	1.8	$0.8-1.2 \times 10^{-3}\text{km}^{-1}$
Market volatility	-0.67	0.28	2.4	0.12-0.18
Form	$\dot{\rho} = \mu_{\max}(1 - e^{-\beta z})\rho, \quad \dot{\theta} = \omega, \quad \dot{z} = \omega - \mu_{\max} \rho^2 e^{-\beta z}$			

All six systems share the same contact normal form. The parameters differ; the structure is identical. This table is the Mini-Beast in one page.

Part III

The 14-Week Program

PEDAGOGY: CEFR $\rightarrow$ TO/TOGT

The Correspondence

The 14-week program maps CEFR language levels to TOGT structural levels, ensuring that cognitive demand never exceeds linguistic capacity.

CEFR	TOGT	Task type	Output	Operator
A1	1	Match, point, choose	1 word	C only
A2	2	Complete, label	1-2 sentences	$C \rightarrow K$
B1	3	Explain, compare	3-4 sentences	$C \rightarrow K \rightarrow$
B2	4	Justify, build	Paragraph	Full G
C1	5	Analyze, critique	Essay	$G + \text{norm}$
C2	6	Synthesize, conjecture	Full argument	Category
D1	7	Original research	Publication	Codex

The 14-Week Structure

Weeks 1-2 The operator sequence in daily life. $C \rightarrow K \rightarrow F \rightarrow U$ in biology, weather, cooking. A2 tasks.

Weeks 3-4 Biological instantiations. HPA, circadian, neural rhythms. B1 tasks.

Weeks 5-6 Plasma reconnection. The X-point, the fold, the new topology. B1-B2.

Weeks 7-8 Market volatility. Price trajectories, regime shifts, κ^* . B2 tasks.

Weeks 9-10 Neural embedding geometry. The coherence bridge. B2-C1.

Weeks 11-12 The Coherence Bridge Theorem. All six orbits as one system. C1.

Weeks 13-14 Student research contributions. Nobel Contributions. The hive mind. C1-C2.

The Three Seed Sentences

A student who completes this program can say — and defend with the full mathematical apparatus:

1. *“In dusty plasmas, magnetic reconnection is a precise generative transition governed by $C \rightarrow K \rightarrow F \rightarrow U$. The dm^3 contact geometry predicts the*

exact reconnection rate and fractal current-sheet structure that standard MHD cannot resolve.”

2. *“Market price trajectories are generative transitions on a volatility manifold. The dm^3 operator algebra predicts regime shifts before they happen, turning crashes into predictable unfolding events rather than black swans.”*
3. *“Neural oscillations, plasma reconnection, market regime shifts, and biological allostasis are not analogies — they are different orbits of the same dm^3 system. The $G^6:33$ crystal is the common substrate. Coherence across domains is exact mathematical identity.”*

These three sentences carry the full weight of the 18,000-page Codex without requiring the private pages.

The seed is planted. The fixed point exists. It is theirs.

ABOUT THE SERIES

Book	Title	ISBN
1	Applications of GOMC Science (Vols. I-III)	979
2	Contact Realization	979
3	The Mini-Beast (this volume)	979
<i>Complete Series</i>	Principia Orthogona: The Complete Series	979
<i>IMPA Edition</i>	Complete Series IMPA.br	979

ABOUT THE AUTHOR

Pablo Nogueira Grossi is an independent researcher and founder of G6 LLC, a mathematical research organization based in Newark, New Jersey. He developed the GOMC framework over twenty-five years, beginning in Brasília in the year 2000, in parallel with a career as an ESL teacher. He is the creator of the TOGT/TO/TOGT framework and the architect of the AXLE formal verification environment in Lean 4.

ORCID: 0009-0000-6496-2186 · Zenodo: 10.5281/zenodo.19117400

eBook only — a living document.

This pilot expands as new instantiations are verified.

G6 LLC · Newark, New Jersey · 2026